

Reviewer Pack — Minimal Rank of Tropicalized Matroid Representations vs Mono...

Entry 8e27f9e14a1a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Tropicalized Matroid Representations vs Monotone k-CLIQUE Circuit Size

Entry ID: 8e27f9e14a1a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 07:44:08 UTC

1. Conjecture

Field A (mathematical branch): Matroid Theory **Field B** (complexity object): Complexity Theory: Monotone Circuit Complexity

Statement:

[“For every monotone DNF formula F representing the k -CLIQUE problem, there exists a representation of its associated matroid M in the tropical semiring such that the minimal rank of M ’s representation is lower-bounded by 2^k .”]

Rationale (proposer’s reasoning):

[‘Matroids provide a combinatorial framework to study independence and dependence relations. Tropicalization, a technique from tropical geometry, has been used to analyze computational problems in complexity theory. The conjecture posits that the minimal rank of a matroid representation could serve as a measure for monotone circuit complexity.’]

Taxonomy category: MONOTONE_CLIQUE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 277f51e1bd6f9c0e

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Spearman’s rank correlation coefficient between minimal ranks of tropicalized matroid representations and monotone k -CLIQUE circuit sizes is ≥ 0.8 , and all seeds produce a correlation coefficient ≤ 0.7 for falsification.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - tropicalization Matroid Theory monotone k-CLIQUE Circuit Complexity - minimal rank tropical semiring Matroid Theory monotone k-CLIQUE - Matroid Theory representation tropical semiring lower bound 2^k monotone k-CLIQUE

Top relevant hits considered: - [http://arxiv.org/abs/1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [http://arxiv.org/abs/1102.2932v2] Applications of Monotone Rank to Complexity Theory - [http://arxiv.org/abs/2405.03576v2] Tropical vector bundles and matroids - [http://arxiv.org/abs/1009.4750v3] Triangulations of $\Delta_{n-1} \times \Delta_{d-1}$ and Tropical Oriented Matroids - [http://arxiv.org/abs/1503.01392v2] Valuations of Semirings - [http://arxiv.org/abs/1603.02493v4] Clifford theory for glider representations - [http://arxiv.org/abs/1406.3065v2] Lower Bounds for Tropical Circuits and Dynamic Programs

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction
import itertools

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_k_clique_cnf(k, n):
        if k == 1:
            return [[random.randint(1, n)] for _ in range(n)]
        elif k == n:
            return [list(range(1, n + 1))]
        else:
            edges = set(itertools.combinations(range(1, n + 1), 2))
            while len(edges) > (n * (n - 1)) // 2 - k * (k - 1):
                u, v = random.sample(range(1, n + 1), 2)
                if (u, v) in edges:
                    edges.remove((u, v))
            return [[i for i in range(1, n + 1) if (i, j) not in edges] for j in range(1, n + 1)]

    def tropical_rank(matrix):
        m, n = len(matrix), len(matrix[0])
        rank = 0
        for col in range(n):
            pivot_row = None
```

```

        for row in range(m):
            if matrix[row][col] != float('-inf'):
                if pivot_row is None:
                    pivot_row = row
                else:
                    ratio = Fraction(matrix[pivot_row][col], matrix[row][col])
                    for j in range(n):
                        matrix[row][j] -= ratio * matrix[pivot_row][j]
            if pivot_row is not None:
                rank += 1
        return rank

n = random.randint(5, 40)
k = random.randint(2, min(n - 1, 3))
cnf = generate_k_clique_cnf(k, n)

tropical_matrix = [[float('-inf')] * n for _ in range(n)]
for clause in cnf:
    for literal in clause:
        if literal > 0:
            tropical_matrix[literal - 1][literal - 1] = max(tropical_matrix[literal - 1][literal - 1], 1)
        else:
            tropical_matrix[-literal - 1][-literal - 1] = max(tropical_matrix[-literal - 1][-literal - 1], 1)

rank = tropical_rank(tropical_matrix)

return {
    "metric_name": "tropical_rank",
    "metric_value": rank,
    "instances_tested": 1,
    "conjecture_holds": rank >= 2 ** k,
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] if len(sys.argv) > 1 else [593, 631, 677, 727, 773]
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if len(results) == 0:
        print("RESULT: INCONCLUSIVE no_results")
    else:
        mean_rank = sum(result["metric_value"] for result in results) / len(results)
        support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

        if support_fraction >= 0.8:
            print(f"RESULT: SUPPORTED mean={mean_rank} std=NA support_fraction={support_fraction}")
        elif any(not result["conjecture_holds"] for result in results):
            first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
            print(f"RESULT: FALSIFIED counterexample=\"first failing seed\" first_failing_seed={first_failing_seed}")
        else:
            print("RESULT: INCONCLUSIVE no_support")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
rank', 'metric_value': 21, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 27, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 27, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 21, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 18, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 15, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 7, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 25, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 23, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 9, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 38, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 36, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 19, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_rank', 'metric_value': 17, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
RESULT: SUPPORTED mean=20.766666666666666 std=NA support_fraction=1.0
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The empirical test only includes a very small sample size of $n \leq 15$, which is insufficient to draw conclusions about the conjecture’s validity. The metric does not scale trivially with n , but the lack of diversity in the tested instances raises concerns about generalizability.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge | original judge: The test results indicate that the conjecture holds for all tested instances, with a mean tropical rank of 20.77 and standard deviation not available | next: Further investigation is needed to confirm the generalizability of these results. We should increase the sample size and test a wider variety of instances to ensure that the observed trend holds for larger problem sizes.

11. Audit log (LLM calls)

Total LLM calls: 13

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12424
2	propose	ollama_remote	glm4:latest	0	0	11182
3	propose	ollama_remote	glm4:latest	0	0	11809
4	propose	ollama_remote	glm4:latest	0	0	9648
5	preregistration	ollama_remote	glm4:latest	0	0	5495
6	novelty	ollama_remote	glm4:latest	0	0	4708
7	novelty	ollama_remote	glm4:latest	0	0	5633

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14239
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9979
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7861
11	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10382
12	critic	ollama_remote	glm4:latest	0	0	12406
13	judge	ollama_remote	glm4:latest	0	0	5981

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 121747 ms total latency. Provider mix: {'ollama_remote': 13}

(full prompt+response transcripts available in *research/audit/8e27f9e14a1a.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/8e27f9e14a1a.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/8e27f9e14a1a.tar.gz* (if generated)